

Correction: Properties of Weighted-Average (Theorem 4.7)

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Through personal communication from Daniel Blankenburg, I was notified that the second inequality in the claim of Theorem 4.7 (and also the claim in Theorem 1 of [HCB11]) does not hold. As a counterexample, he provided $\mathbf{x} = (10, 10 - \ln 2, 10 - \ln 2, 0)^T$. The issue in the proof is the final inequality on page 23, which is written as “ $\leq$ ” when it should have been “ $\geq$ ”.

However, the weaker claim that

$$\max(\mathbf{x}) - \gamma W\left(\frac{n-1}{e}\right) \leq \text{WA}_\gamma(\mathbf{x}) \leq \max(\mathbf{x}) \quad \forall \mathbf{x} \in \mathbb{R}^n \quad (1)$$

still holds, as we will prove below. This also means Theorem 4.8 holds as claimed.

The upper bound is clear from the definition. For the lower bound consider that $\text{WA}_\gamma(\mathbf{x}) - \max(\mathbf{x}) = \text{WA}_\gamma(\mathbf{x} - \max(\mathbf{x})\mathbf{1})$, so that it suffices to show that $\text{WA}_\gamma(\mathbf{x}) \geq -\gamma W(\frac{n-1}{e})$ holds whenever at least one entry is zero. Assume without loss of generality that the final entry x_n is equal to zero.

By Theorem 4.9 the gradient of WA_γ is given by

$$\nabla \text{WA}_\gamma(\mathbf{x}) = \sigma(\mathbf{x}/\gamma) \odot \left(\mathbf{1} + \mathbf{x}/\gamma - \frac{1}{\gamma} \text{WA}_\gamma(\mathbf{x})\mathbf{1} \right). \quad (2)$$

Therefore, if $a_0: \mathbb{R}^{n-1} \rightarrow \mathbb{R}^n$ maps $(y_1, \dots, y_{n-1})^T$ to $(y_1, \dots, y_{n-1}, 0)^T$, then stationarity for $\text{WA}_\gamma(a_0(\mathbf{y}))$ for $\mathbf{y} \in \mathbb{R}^{n-1}$ means

$$\sigma(a_0(\mathbf{y})/\gamma)_i \left(1 + y_i/\gamma - \frac{1}{\gamma} \text{WA}_\gamma(a_0(\mathbf{y})) \right) = 0 \quad (3)$$

for all $1 \leq i \leq n-1$. Since the output of softmax is positive in each component, this implies that the second term is zero and so that

$$y_i = \text{WA}_\gamma(a_0(\mathbf{y})) - \gamma. \quad (4)$$

Therefore any stationary point of $\text{WA}_\gamma \circ a_0$ has the form $(y, \dots, y, 0)$ for some $y \in \mathbb{R}$. For such points the function value for WA is

$$\text{WA}_\gamma(y, \dots, y, 0) = \frac{(n-1)ye^{y/\gamma}}{(n-1)e^{y/\gamma} + 1}. \quad (5)$$

Combining this with equation 4 we conclude that the stationary points of $\text{WA}_\gamma \circ a_0$ are exactly the points $\mathbf{x} = (y, \dots, y, 0)$ such that

$$y = \frac{(n-1)ye^{y/\gamma}}{(n-1)e^{y/\gamma} + 1} - \gamma \quad (6)$$

Rearranging gives

$$0 = (n-1)ye^{y/\gamma} - (y+\gamma)((n-1)e^{y/\gamma} + 1) \quad (7)$$

$$= -\gamma(n-1)e^{y/\gamma} - (y+\gamma) \quad (8)$$

$$= \gamma e^{y/\gamma+1}((n-1)e^{-1} - (y/\gamma + 1)e^{-(y/\gamma+1)}) \quad (9)$$

Therefore, $-(y/\gamma + 1) = W(\frac{n-1}{e})$ by definition of the Lambert W function and using equation 4, $\text{WA}_\gamma(\mathbf{x}) = -\gamma W(\frac{n-1}{e})$ for any stationary point of $\text{WA}_\gamma \circ a_0$.

It remains to show that this is the global minimum of $\text{WA}_\gamma(a_0(\mathbf{y}))$. The argument is by induction. Clearly, for $n = 1$, since the only input to WA_γ is fixed at zero, the minimum is $0 = -\gamma W(\frac{0}{e})$. Now assume that $\text{WA}_\gamma(a_0(\mathbf{y})) \geq -\gamma W(\frac{k-1}{e})$ for all $\mathbf{y} \in \mathbb{R}^{k-1}$ for some $k \geq 1$ and consider the case for one dimension higher. For this purpose let $\mathbf{y} \in \mathbb{R}^k$ and assume that $\text{WA}_\gamma(a_0(\mathbf{y})) < 0$ for now. In this case, we know that $\sum_{i=1}^k y_i e^{y_i/\gamma} < 0$ and for any $1 \leq j \leq k$ we have

$$-\text{WA}_\gamma(a_0(\mathbf{y})) = \frac{-\sum_{i=1}^k y_i e^{y_i/\gamma}}{\sum_{i=1}^k e^{y_i/\gamma} + 1} < \frac{-\sum_{i=1}^k y_i e^{y_i/\gamma}}{\sum_{i=1, i \neq j}^k e^{y_i/\gamma} + 1} \quad (10)$$

$$= \frac{-y_j e^{y_j/\gamma}}{\sum_{i=1, i \neq j}^k e^{y_i/\gamma} + 1} + \frac{-\sum_{i=1, i \neq j}^k y_i e^{y_i/\gamma}}{\sum_{i=1, i \neq j}^k e^{y_i/\gamma} + 1} \quad (11)$$

$$\leq \frac{-y_j e^{y_j/\gamma}}{\sum_{i=1, i \neq j}^k e^{y_i/\gamma} + 1} + \gamma W(\frac{k-1}{e}) \quad (12)$$

$$\leq \max\{0, -y_k e^{y_k/\gamma}\} + \gamma W(\frac{k-1}{e}) \quad (13)$$

where the second inequality follows from our induction hypothesis. Considering $z \mapsto \max\{0, -ze^{z/\gamma}\}$ we see that the function converges to zero for $z \rightarrow \pm\infty$. Thus, there is an $M > 0$ such that $\max\{0, -ze^{z/\gamma}\} < \gamma W(\frac{k}{e}) - \gamma W(\frac{k-1}{e})$ for all $z \notin (-M, M)$. By the derivation above, this means that $\text{WA}_\gamma(a_0(\mathbf{y})) > -\gamma W(\frac{k}{e})$ for any $\mathbf{y} \notin (-M, M)^k$. This also applies to all $\mathbf{y}$ with $\text{WA}_\gamma(a_0(\mathbf{y})) \geq 0$ that we ignored previously. Since $\text{WA}_\gamma \circ a_0$ is continuous it attains its minimum on $[-M, M]^k$. As already shown, this minimum value is at least $-\gamma W(\frac{k}{e})$, which means the minimum is not attained at the boundary but in the interior of $[-M, M]^k$. As such, the point at which it is attained must be a stationary point and by our previous analysis all stationary points have the function value $-\gamma W(\frac{k}{e})$. This shows that $\text{WA}_\gamma \circ a_0$ is lower bounded by $-\gamma W(\frac{k}{e})$ on $[-M, M]^k$ and outside of it for $n = k + 1$. By induction the claim follows for all n .